

The Enumerative Combinatorics of Genome Inversions

Frank Burkhart

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Joint work with Dana Ernst, Alexander Stewart, and others.

Cabbage to Turnip

An *inversion* is a chromosome rearrangement where a subsequence of the chromosome is "inverted."

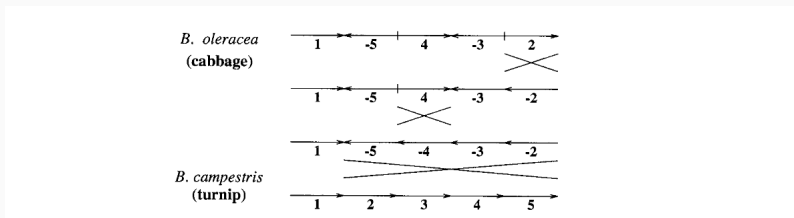


Figure 1: Hannenhalli & Pevzner, *Transforming cabbage to turnip*. Five genes inverted three times to transform cabbage mitochondrial genome into turnip mitochondrial genome.

Signed permutations and reversal model

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Definition A *reversal* of a signed permutation is the action of swapping the order of a consecutive subsequence of numbers and changing the sign of each number in the subsequence.

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Example signed permutation $\pi \in S_4^\pm$, $\pi = [-2, -1, 3, -4]$. Let's use reversals to sort this permutation into the identity permutation.

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$$\begin{aligned}\pi &= [\underline{-2}, \underline{-1}, 3, -4] \\ &\xrightarrow{\delta_{1,2}} [1, 2, 3, \underline{-4}] \\ &\xrightarrow{\delta_{4,4}} [1, 2, 3, 4]\end{aligned}$$

Then we say that π has *reversal distance* 2. We denote this by $d(\pi) = 2$.

More on reversal distance

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Goal: We want to find closed form solutions for each $r_{n,k}$

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$$R_{2,0} = \{[1, 2]\}$$

$$R_{2,1} = \{[-1, 2], [1, -2], [-2, -1]\}$$

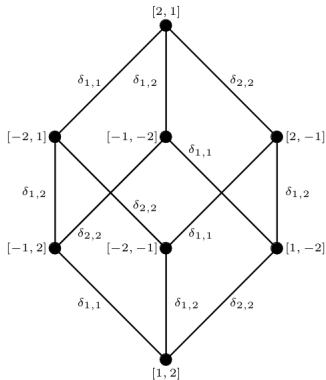
$$R_{2,2} = \{[-1, -2], [-2, 1], [2, -1]\}$$

$$R_{2,3} = \{[2, 1]\}$$

Then we see $r_{2,0} = 1$, $r_{2,1} = 3$, $r_{2,2} = 3$ and $r_{2,3} = 1$

Reversal poset

Another way to think about the problem is as a counting rows in a partially ordered set.



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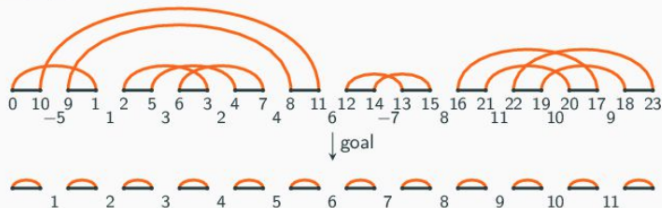
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We will need another tool to get beyond these first few rows.

Breakpoint diagrams

Each signed permutation has an associated *breakpoint diagram*.

Here is one for $\pi = [-5, 1, 3, 2, 4, 6, -7, 8, 11, 10, 9] \in S_{11}^{\pm}$



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Theorem (Hannenhalli, Pevzner) The reversal distance of a signed permutation $\pi \in S_n^\pm$ is given by,

$$d(\pi) = n + 1 - c(\pi) + h(\pi) + f(\pi)$$

where $c(\pi)$, $h(\pi)$ and $f(\pi)$ are all "characteristics" of the breakpoint diagram of π .

Top row of the poset

The maximum reversal distance of permutations in $S_n^\pm$ is given by,

$$d_k^{\max} = \begin{cases} n & \text{if } n = 1, 3 \\ n + 1 & \text{otherwise} \end{cases}$$

Theorem (Awik, Ernst): For $n \neq 1, 3$, we have

$$r_{n,n+1} = \sum_{(\alpha_1, \dots, \alpha_k) \in \mathcal{C}_1(n+1)} \left(\prod_{i=1}^k \frac{2(\alpha_i - 1)!}{\alpha_i + 1} \right) \cdot \begin{cases} \alpha_i & k \neq 1 \\ 1 & k = 1 \end{cases}$$

Generalized formula

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Theorem (Burkhart, Ernst, Stewart):

$$r_{n,k} = \sum_{i=1}^{n+1} a_{i,k} \binom{n+1}{i+1}$$

Where $a_{i,k}$ is the number of signed permutations in $S_i^{\pm}$ with no "trivial cycles".

Open problems

- Figure out how to count these trivial cycles
- Generate more data to verify results
- Does any of this help anyone???